



第八章 多元函数微分学

n 元函数的定义: 设 $D \subset \mathbb{R}^n, D \neq \emptyset$

如果存在一个对应法则 f , 对每一个 P

$(x_1, x_2, \dots, x_n) \in D$, 都有唯一实数 y

与之对应. 称 $f: \forall P \in D \rightarrow y$ 是 D 上的

n 元函数. 记作 $y = f(P), P \in D$.

或 $y = f(x_1, x_2, \dots, x_n) P(x_1, x_2, \dots, x_n) \in D$.

称 $x_1, x_2, \dots, x_n$ 为自变量, y 称为因变量.

D 称为 f 的定义域.

$\{f(P): P \in D \subset \mathbb{R}^n\} \triangleq R(f)$ 称为 f 的值域.

称 $\{(x_1, x_2, \dots, x_n, y): y = f(x_1, x_2, \dots, x_n)$

其中 $(x_1, x_2, \dots, x_n) \in D \subset \mathbb{R}^n\} \subset \mathbb{R}^{n+1}$

称为 n 元函数的图形.

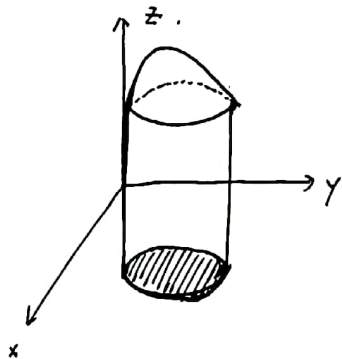
当 $n=1$ 时称为一元函数, $n>1$ 时称为多元函数.

以后经常研究二元函数. $z = f(x, y)$

$(x, y) \in D \subset \mathbb{R}^2 \quad \{(x, y, z): z = f(x, y), (x, y) \in D \subset \mathbb{R}^2\}$

$\subset \mathbb{R}^3$ 称为二元函数的图形. 一般是三维空间

的一张曲面.



特例: $z = \sqrt{1-x^2-y^2}$

解. 要使表达式有意义

$$-(x^2+y^2) \geq 0 \Leftrightarrow x^2+y^2 \leq 0$$

$$\Rightarrow x^2+y^2=0 \Rightarrow x=0, y=0.$$

定义域是 $\{(0, 0)\}$.

图形 $\{(0, 0, 0)\}$ 是空间一点

三元函数

$$u = f(x, y, z) \quad (x, y, z) \in D \subset \mathbb{R}^3$$

它的图形 $\{(x, y, z, u): u = f(x, y, z)$

$(x, y, z) \in D \subset \mathbb{R}^3\} \subset \mathbb{R}^4$ 称为超曲面.

定义域的求法 (以二元为例)

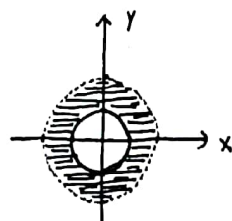
1. $z = f(x, y)$ 是由数学表达式给出.

它的定义域就是使表达式有意义的全体平面上的点组成的集合.

例: $z = \sqrt{x^2+y^2-1} + \ln(4-x^2-y^2)$ 求定义域.

$$\begin{cases} x^2+y^2-1 \geq 0 \\ 4-x^2-y^2 > 0 \end{cases} \Leftrightarrow \begin{cases} x^2+y^2 \geq 1 \\ x^2+y^2 < 4 \end{cases}$$

定义域为 $\{(x, y): \begin{cases} x^2+y^2 \geq 1 \\ x^2+y^2 < 4 \end{cases}\}$



包括 $\rightarrow$ 实线

不包括 $\rightarrow$ 虚线.

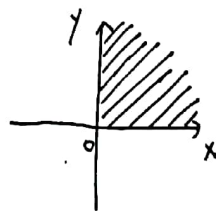
2. 如果函数用数学表达式表示, 还涉及到实际问题.

例: $S = xy$

其中 x, y 为矩形边长, S 为矩形面积.

求定义域.

定义域为 $\{(x, y): x \geq 0, y \geq 0\}$





3. 例 $z = x^2 + y^2$, 且满足
 $\{(x, y): x^2 + y^2 \leq 1\}$ 则它的定义域
就是 $\{(x, y): x^2 + y^2 \leq 1\}$

二. 平面点集.

设 $P_0(x_0, y_0) \in \mathbb{R}^2$

$\{P: d(P, P_0) < \delta \text{ (}\delta \text{ 是某正数, } P \in \mathbb{R}^2)\}$

$\triangleq U(P_0, \delta)$ 称为 P_0 的 δ 邻域.

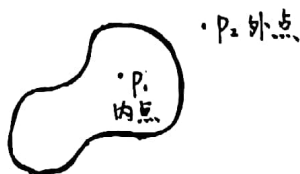
$\{P: 0 < d(P, P_0) < \delta \text{ (}\delta \text{ 是某正数, } P \in \mathbb{R}^2)\}$

$\triangleq \dot{U}(P_0, \delta)$ 称为 P_0 的 δ 空心邻域.

设 $P_0(x_0, y_0) \in \mathbb{R}^2, E \subset \mathbb{R}^2$

1. 内点: 如果 $\exists \delta > 0$, 使 $U(P_0, \delta) \subset E$
称 P_0 为 E 的内点, E 中全体内点组成的
集合称为 E 的内部. 记作 $\text{int } E$

2. 外点: 如果 $\exists \delta > 0$, 使 $U(P_0, \delta) \cap E = \emptyset$
称 P_0 为 E 的外点, 且 $P_0 \notin E$



3. 边界点: $\forall \delta > 0, U(P_0, \delta)$ 中既有 E 中的点,
又含有不在 E 中的点, 称 P_0 为边界点.
此时 P_0 也可能属于 E , 也可能不属于 E .



$$E = D \cup \{P_1\}$$

P_1 是 E 的边界点 $P_1 \notin E$

P_2 是 E 的边界点 $P_2 \in E$

P_3 是 E 的边界点, $P_3 \in E$ (孤立点)

E 全体边界点组成的集合称为 E 的边界.

记为 ∂E

4. 若 $\text{int } E = E$, 称 E 为开集.

5. 若 $\mathbb{R}^2 - E$ 为开集, 称 E 为闭集.

6. 若对 E 中的任意两点 P_1, P_2 都可以用
属于 E 的折线段连接起来, 则称 E 是
连通的.



7. 若 E 是开集且连通的
称 E 为开区域

8. 开区域加上全体边界点组成的集合
称为闭区域

9. 开区域 - 闭区域, 或开区域加上部分
边界, 并统称为区域.

10. 设 O 为平面上一点, 若 $\exists \delta > 0$ 使
 $E \subset U(O, \delta)$, 称 E 为有界集. 否则无界集
或若存在常数 $M > 0$ 使
 $\sup\{d(P_1, P_2) | P_1, P_2 \in E\} \leq M$ 则 E 为有界集.
上确界 称为 E 的直径





$\{(x, y): x^2 + y^2 < 1\}$, ϕ , R^2 均为开集, 开区域

$\{(x, y): x^2 + y^2 \leq 1\}$, ϕ , R^2 均为闭集, 闭区域

$\{(x, y): x^2 + y^2 \leq 1\}$ 是有界闭区域.

R^2 不是有界闭区域.

三. 多元函数的极限. (以二元函数为例)

定义: 设 $P_0(x_0, y_0) \in R^2$, $\exists \delta_0 > 0$. $z = f(x, y)$

在 $\dot{U}(P_0, \delta_0)$ 内有定义. A 为一个确定的常数.

对于 $\forall \varepsilon > 0$, $\exists \delta > 0$ (且 $\delta \leq \delta_0$) 当 $0 < d(P, P_0) < \delta$

即 $0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$ 或 $P \in \dot{U}(P_0, \delta)$ 时

都有 $|f(x, y) - A| < \varepsilon$ 成立. 称 $P(x, y)$ 趋于

$P_0(x_0, y_0)$ 时 $f(x, y)$ 的极限是 A . 记作:

$$\lim_{P \rightarrow P_0} f(P) = A \text{ 或 } \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y) = A \text{ 或 } \lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = A$$

$$\text{或 } f(x, y) \rightarrow A \quad \left(\begin{matrix} x \rightarrow x_0 \\ y \rightarrow y_0 \end{matrix} \right)$$

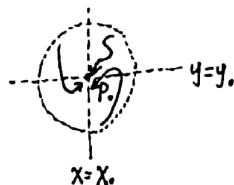
$$\text{例: } \lim_{x \rightarrow x_0} f(x) = A \text{ (常)} \Leftrightarrow f(x) = A + \alpha(x)$$

$$\text{其中 } \lim_{x \rightarrow x_0} \alpha(x) = 0$$

在二元函数中, 有.

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y) = A \text{ (常)} \Leftrightarrow f(x, y) = A + \alpha(x, y)$$

$$\text{其中 } \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} \alpha(x, y) = 0$$



$$P(x, y) \rightarrow P_0(x_0, y_0) \quad P \neq P_0, (x, y) \neq (x_0, y_0)$$

要求不能同时有 $x = x_0, y = y_0$. 但是可以

$x = x_0, y \neq y_0$ 或 $y = y_0, x \neq x_0$.

判断二元函数极限不存在的方法:

在 $\dot{U}(P_0, \delta)$ 中, 选择两个不同路径

$P \rightarrow P_0$, 而 $z = f(x, y)$ 沿着这两个路径

极限存在, 但不相等, 则:

$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y)$ 必定不存在 (用的多)

在 $\dot{U}(P_0, \delta)$ 中取一路径 $P \rightarrow P_0$

$f(x, y)$ 沿这个路径极限不存在.

则 $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y)$ 不存在. (用的少)

$$\begin{aligned} \text{例: } \lim_{\substack{x \rightarrow 0 \\ y \rightarrow a}} \frac{\sin xy}{y} \quad (a \neq 0) &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow a}} \frac{\sin xy}{y} \\ &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow a}} \frac{\sin xy}{xy} \cdot x = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow a}} \frac{\sin xy}{xy} \cdot x \\ &= 1 \cdot 0 = 0 \end{aligned}$$

在求极限的时候, 有一类极限.

$$\lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) \quad \lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x, y)$$

$$(x \neq x_0, y \neq y_0)$$

这两类本质上是二次求一元函数的极限, 称为二次极限或累次极限. 与我们前面讲的重极限

$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y)$ 是不同的.

(x, y 不同时为 x_0, y_0 . 即 $(x, y) \neq (x_0, y_0)$)

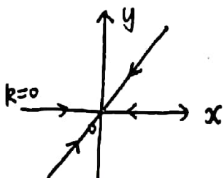




例: $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{2xy}{x^2+y^2}$ (齐次 $\rightarrow y=kx$)

解: $\lim_{\substack{x \rightarrow 0 \\ y=kx}} \frac{2xy}{x^2+y^2} = \lim_{x \rightarrow 0} \frac{2x \cdot kx}{x^2+k^2x^2} = \frac{2k}{1+k^2}$

$= \begin{cases} 0 & k=0 \\ 1 & k=1 \end{cases}$



知 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{2xy}{x^2+y^2}$ 不存在

注: 分子、分母次数同的单项式的加、减
趋于原点时一般极限不存在

例 $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} \frac{2xy}{x^2+y^2} = \lim_{y \rightarrow 0} \frac{0}{y^2} = 0$ $\downarrow \begin{matrix} x \\ y \end{matrix}$

利用轮换 (同理)

$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} \frac{2yx}{y^2+x^2} = \lim_{x \rightarrow 0} 0 = 0$

例: $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x \sin \frac{1}{y} + y \sin \frac{1}{x}) = 0$

$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} (x \sin \frac{1}{y} + y \sin \frac{1}{x})$ 不存在

同理, $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} (y \sin \frac{1}{x} + x \sin \frac{1}{y})$ 不存在

定理: 若 $\lim_{x \rightarrow x_0} f(x, y)$, $\lim_{y \rightarrow y_0} f(x, y)$, $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y)$

都存在, 则三者相等。

逆否: 若两个累次极限存在不相等, 则
= 重极限不存在。

推论: 若 $\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x, y) = A$ (常) $\lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) = B$ (常)

且 $A \neq B$ 则 $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y)$ 不存在。

例: $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2-y^2}{x^2+y^2}$

解: $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} \frac{x^2-y^2}{x^2+y^2} = -1$

$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} \frac{x^2-y^2}{x^2+y^2} = 1$

$-1 \neq 1$ 知 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2-y^2}{x^2+y^2}$ 不存在

或

$\lim_{\substack{x \rightarrow 0 \\ y=kx}} \frac{x^2-y^2}{x^2+y^2} = \lim_{x \rightarrow 0} \frac{x^2-k^2x^2}{x^2+k^2x^2}$

$= \frac{1-k^2}{1+k^2} \begin{cases} 0 & k=1 \\ 1 & k=0 \end{cases}$

例: $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2-y^4}{x^2+y^4}$

解: $\lim_{\substack{y \rightarrow 0 \\ x=ky}} \frac{k^2y^4-y^4}{k^2y^4+y^4} = \frac{k^2-1}{k^2+1} \begin{cases} 0 & k=1 \\ -1 & k=0 \end{cases}$

故原函数极限不存在。

多元函数的连续 (以二元为例)

设 $P_0(x_0, y_0) \in \mathbb{R}^2$

定义: 设 $f(x, y)$ 在 $U(P_0, \delta)$ ($\delta > 0$) 内有定义

且 $\lim_{P \rightarrow P_0} f(P) = f(P_0)$ 或 $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y) = f(x_0, y_0)$

称 $z = f(x, y)$ 在 $P_0(x_0, y_0)$ 处连续。

否则称 $f(x, y)$ 在 P_0 处不连续, 称 $P_0(x_0, y_0)$ 为间断点





设 G 为一个区域. 若 $\forall P(x, y) \in G, z = f(x, y)$ 在 $P(x, y)$ 处都连续. 则称 $z = f(x, y)$ 在区域 G 上连续. 此时 $z = f(x, y)$ 的图形是 G 上的一张连续的曲面.

有界闭区域 G 上连续函数的性质:

定理: 设 G 为有界闭区域, 且 $f(P)$ 在 G 上连续. 则 $f(P)$ 在 G 上一定能取到最大值 M 与最小值 m . 且可以取到 $[m, M]$ 上的每一个值. 即 $R(f) = [m, M]$.

$$\forall P_1, P_2 \in G, P_1 \neq P_2 \text{ 若 } f(P_1)f(P_2) < 0, \\ \exists P^* \in G, \text{ 使 } f(P^*) = 0$$

如果 $f(x, y)$ 在 $P_0(x_0, y_0)$ 处连续 $\Leftrightarrow$

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y) = f(x_0, y_0) \Leftrightarrow \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} [f(x, y) - f(x_0, y_0)] = 0$$

$$\text{令 } x - x_0 = \Delta x \text{ 即 } x = x_0 + \Delta x \quad \Leftrightarrow \\ y - y_0 = \Delta y \text{ 即 } y = y_0 + \Delta y$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} [f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)] = 0$$

令 $f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = \Delta z$ 称为 $z = f(x, y)$ 在 $P_0(x_0, y_0)$ 处函数值的全增量.

$$\Leftrightarrow \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \Delta z = 0 \text{ 称 } z = f(x, y) \text{ 在 } P_0(x_0, y_0) \text{ 处连续}$$

令 $f(x_0 + \Delta x, y_0) - f(x_0, y_0) = \Delta xz$ 称为 $z = f(x, y)$ 在 $P_0(x_0, y_0)$ 处函数值对 x 的偏增量.

令 $f(x_0, y_0 + \Delta y) - f(x_0, y_0) = \Delta yz$ 称为 $z = f(x, y)$ 在 $P_0(x_0, y_0)$ 处函数值对 y 的偏增量.

定理: 初等多元函数在定义域区域上的每一点处都连续

定义: 由 x 为自变量的一元基本初等函数或以 y 为自变量的一元基本初等函数经过有限次的四则运算或复合运算得到的二元函数, 称为初等二元函数.

$$\text{例: } \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{1 + xy + e^{\sin(x^2 + y^2)}}{1 + \sqrt{1 + \sin xy + x^2 + y^2}} \\ = \frac{1 + 0 + e^0}{1 + \sqrt{1}} \\ = 1$$

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & x^2 + y^2 > 1 \\ x^2 + y^2 & x^2 + y^2 \leq 1 \end{cases} \text{ 称为分片函数 (分块)}$$

$$f(x, y) = \begin{cases} x^2 + y^2 & (x, y) \neq (0, 0) \\ 1 & (x, y) = (0, 0) \end{cases} \text{ 也为分片函数}$$

多元函数的偏导数 (以二元为例)

例: $z = f(x, y)$, 所谓 z 对 x 的偏导.

记作 $\frac{\partial z}{\partial x}$, 即此时把 y 视为常数.

本质上看成 z 是 x 的一元函数. 然后用一元函数的求导公式 z 对 x 求导 (视 y 为常数)

同理, $\frac{\partial z}{\partial y}$ 指 z 对 y 偏导 (视 x 为常数) 本质上是 z 对 y 求导.





例: $z = x^y$, 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

解: $\frac{\partial z}{\partial x} = yx^{y-1}$

$$\frac{\partial z}{\partial y} = x^y \ln x$$

例: $u = x^{y^2}$, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial u}{\partial z}$

注: $x^{y^2} \neq (x^y)^2 = x^{2y}$

解: $\frac{\partial u}{\partial x} = y^2 x^{y^2-1}$

$$\frac{\partial u}{\partial y} = x^{y^2} \ln x (y^2)' = 2y x^{y^2} \ln x$$

$$\frac{\partial u}{\partial z} = x^{y^2} \ln x (y^2)'_z = 2y x^{y^2} \ln x y^z \ln y$$

例: $z = (x^2 + y^2)^{xy}$ 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$

解: $z = e^{\ln(x^2+y^2)^{xy}} = e^{xy \ln(x^2+y^2)}$

$$\frac{\partial z}{\partial x} = e^{xy \ln(x^2+y^2)} (y \ln(x^2+y^2) + xy \cdot \frac{2x}{x^2+y^2})$$

$$\text{轮换可知 } \frac{\partial z}{\partial y} = x e^{xy \ln(x^2+y^2)} [\ln(x^2+y^2) + \frac{2y^2}{x^2+y^2}]$$

例: 设时间为 t , 温度为 T , 求

$T = f(x, y, z, t)$, 求在地点 (x_0, y_0, z_0) 处, 在 t_0 时刻, 温度 T 对时间 t 的变化率

解: $T = f(x_0, y_0, z_0, t) \triangleq \varphi(t)$ 求 $\varphi(t)$ 在 $t=t_0$ 时

对 t 的导数. 即 $\lim_{\Delta t \rightarrow 0} \frac{\varphi(t_0 + \Delta t) - \varphi(t_0)}{\Delta t} = \varphi'(t_0)$

$$\Leftrightarrow \lim_{\Delta t \rightarrow 0} \frac{f(x_0, y_0, z_0, t_0 + \Delta t) - f(x_0, y_0, z_0, t_0)}{\Delta t} = \left. \frac{df(x_0, y_0, z_0, t)}{dt} \right|_{t=t_0} = \left. \frac{\partial f(x, y, z, t)}{\partial t} \right|_{\substack{x=x_0 \\ y=y_0 \\ z=z_0 \\ t=t_0}}$$

给出 $z = f(x, y)$ 在 (x_0, y_0) 点, 对 x 的偏导数.

为 $z = f(x, y) \triangleq \varphi(x)$ 在 $x=x_0$ 处对 x 的导数

$$\lim_{\Delta x \rightarrow 0} \frac{\varphi(x_0 + \Delta x) - \varphi(x_0)}{\Delta x} = \varphi'(x_0) \text{ 即}$$

$$\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} \text{ (存在)} \triangleq \frac{\partial f}{\partial x} = \left. \frac{\partial z}{\partial x} \right|_{\substack{x=x_0 \\ y=y_0}}$$

定义: 设 $\exists \delta_0 > 0$, $z = f(x, y)$ 在 $U(P_0, \delta_0)$ 内

有定义, 且 $(x_0 + \Delta x, y_0) \in U(P_0, \delta_0) (P_0(x_0, y_0))$

$$\text{若 } \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x z}{\Delta x}$$

$$\triangleq \lim_{x \rightarrow x_0} \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \text{ (存在) (常数)}$$

$$\triangleq \left. \frac{\partial z}{\partial x} \right|_{\substack{x=x_0 \\ y=y_0}} = z'_x \Big|_{\substack{x=x_0 \\ y=y_0}} = f'_x(x_0, y_0) = \left. \frac{\partial f}{\partial x} \right|_{\substack{x=x_0 \\ y=y_0}}$$

称为 $z = f(x, y)$ 在 (x_0, y_0) 处对 x 的偏导数. 如果极限不存在, 称 $z = f(x, y)$ 在 (x_0, y_0) 处对 x 的偏导数不存在.

定义: 若 $\exists \delta_0 > 0$ 设 $z = f(x, y)$ 在 $U(P_0, \delta_0)$ 内有定义

且 $(x_0, y_0 + \Delta y) \in U(P_0, \delta_0)$. 若 $\lim_{\Delta y \rightarrow 0} \frac{\Delta y z}{\Delta y}$

$$= \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y} \stackrel{y=y_0+\Delta y}{=} \lim_{y \rightarrow y_0} \frac{f(x_0, y) - f(x_0, y_0)}{y - y_0}$$

$$\triangleq \left. \frac{\partial z}{\partial y} \right|_{\substack{x=x_0 \\ y=y_0}} = f'_y(x_0, y_0) = z'_y \Big|_{\substack{x=x_0 \\ y=y_0}} \text{ (存在)}$$

称为 $z = f(x, y)$ 在 (x_0, y_0) 处对 y 的偏导数.





若 $z=f(x,y)$ 在区域 G 上的每一点处对 x, y 的偏导数都存在, 即 $\forall p(x,y) \in G$

$$\lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x} \text{ (存在)} = \frac{\partial z}{\partial x} = z'_x = f'_x(x, y)$$

按照多元函数的定义, $f'_x(x, y)$ 是区域 G 上的 x, y 的函数, 称为 z 对 x 的偏导函数, 简称 z 对 x 的偏导数.

$$\lim_{\Delta y \rightarrow 0} \frac{f(x, y+\Delta y) - f(x, y)}{\Delta y} \text{ (存在)} = \frac{\partial z}{\partial y} = z'_y = f'_y(x, y)$$

是区域 G 上 x, y 的函数, 称为 z 对 y 的偏导函数. 简称为 z 对 y 的偏导数

由前面分析, $f'_x(x_0, y_0) = \left. \frac{df(x, y)}{dx} \right|_{x=x_0}$ 将 (x_0, y_0)

换成 (x, y) 即 $f'_x(x, y) = \left. \frac{df(x, y)}{dx} \right|_{y=y_0}$ (y 视作常数)

同理,

$$f'_y(x_0, y_0) = \left. \frac{df(x, y)}{dy} \right|_{y=y_0} \text{ 或 } f'_y(x, y) = \left. \frac{df(x, y)}{dy} \right|_{x=x_0} \text{ (} x \text{ 视作常数)}$$

总结:

1. 求偏导数 $f'_x(x, y), f'_y(x, y)$

$$(1) f'_x(x, y) = \left. \frac{df(x, y)}{dx} \right|_{y=y_0} \text{ (} y \text{ 为常数)}$$

$$f'_y(x, y) = \left. \frac{df(x, y)}{dy} \right|_{x=x_0} \text{ (} x \text{ 为常数)}$$

(2) 用偏导数定义 (用得少)

2. 求一点的偏导数 $f'_x(x_0, y_0), f'_y(x_0, y_0)$

(1) 求出 $f'_x(x, y), f'_y(x, y)$ 代入 $x=x_0, y=y_0$ (用得多)

$$(2) f'_x(x_0, y_0) = \left. \frac{df(x, y)}{dx} \right|_{x=x_0}, f'_y(x_0, y_0) = \left. \frac{df(x, y)}{dy} \right|_{y=y_0}$$

(3) 用偏导数定义, 用第三种形式.

↳ 主要用于求孤立点.

例: $z = x + y(1 + xy)^{(1+x^2+y^2)^{\frac{1}{2}} \sin(x+y)}$

求 $\frac{\partial z}{\partial x} \Big|_{x=1, y=0}$

解: $\frac{\partial z}{\partial x} \Big|_{x=1, y=0} = \frac{df(x, y)}{dx} \Big|_{x=1} = \frac{dx}{x} \Big|_{x=1} = 1$

例: $f(x, y) = \begin{cases} \frac{2xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$

研究 $f(x, y)$ 在 $(0, 0)$ 处, 两个偏导数是否存在, 在 $(0, 0)$ 处是否连续.

解: $\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x - 0} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0$ (存在)

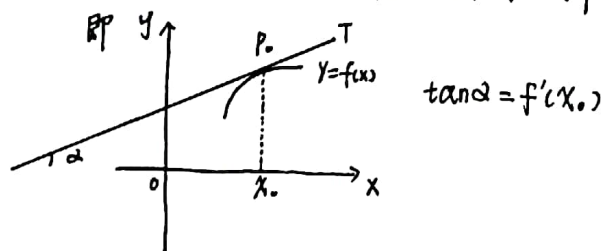
$\therefore f'_x(0, 0)$ 由轮换对称知 $f'_y(0, 0) = 0$

由 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{2xy}{x^2+y^2}$ 不存在.

故在 $(0, 0)$ 处不连续.

注: 在多元函数中 $f(x, y)$ 在一点处偏导数存在与 $f(x, y)$ 在 (x, y) 处连续无关.

在一元函数中, 若 $f'(x_0)$ 存在, 则 $f'(x_0)$ 表示曲线 $y=f(x)$ 在曲线上点 (x_0, y_0) 处切线的斜率

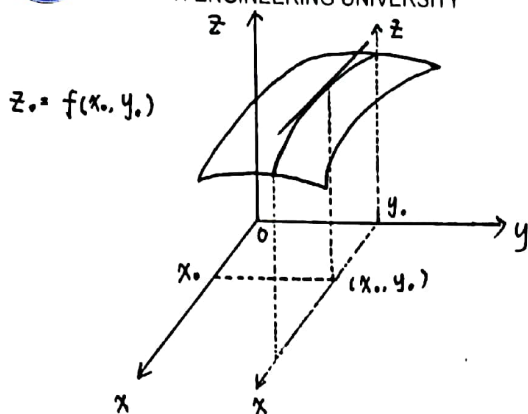


偏导数的几何意义

若 $f'_x(x_0, y_0)$ 存在 $= \left. \frac{df(x, y)}{dx} \right|_{x=x_0}$

由 $\begin{cases} z=f(x, y) \\ y=y_0 \end{cases} \Leftrightarrow \begin{cases} z=f(x, y_0) \\ y=y_0 \end{cases}$





则 $f'_x(x_0, y_0)$ 表示曲线 $\begin{cases} z = f(x, y_0) \\ y = y_0 \end{cases}$ 在 (x_0, y_0, z_0)

处切线与 x 轴正向夹角的正切值。

同理, 若 $f'_y(x_0, y_0)$ 存在, 则 $f'_y(x_0, y_0)$ 表示曲线

$\begin{cases} z = f(x_0, y) \\ x = x_0 \end{cases}$ 在曲线上点 (x_0, y_0, z_0) 处切线
与 y 轴正向夹角的正切值。

例: $z = f(x^2 - 3y^2)$ 且 f 可导, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

解: $\frac{\partial z}{\partial x} = f'(x^2 - 3y^2) \cdot 2x = 2x \cdot f'$

$\frac{\partial z}{\partial y} = f'(x^2 - 3y^2) \cdot (-6y) = -6y \cdot f'$

高阶偏导数

若 $z = f(x, y)$ 的偏导数 $f'_x(x, y), f'_y(x, y)$ 的

偏导数又存在即 $(z'_x)'_x \triangleq z''_{xx} = f''_{xx}(x, y) = \frac{\partial^2 z}{\partial x^2}$

$(z'_x)'_y = z''_{xy} = f''_{xy}(x, y) = \frac{\partial^2 z}{\partial x \partial y}$

$(z'_y)'_x = z''_{yx} = f''_{yx}(x, y) = \frac{\partial^2 z}{\partial y \partial x}$

$(z'_y)'_y = z''_{yy} = f''_{yy}(x, y) = \frac{\partial^2 z}{\partial y^2}$

$(z'_x)'_y = \frac{\partial z'_x}{\partial y} = \frac{\partial}{\partial y} (z'_x) = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial x \partial y}$

$\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}$ 先对 x 偏导, 再对 x 偏导, 再对 y 偏导

例: $z = x^3 + x^2y + y^3$ 求所有二阶偏导数。

解: $\frac{\partial z}{\partial x} = 3x^2 + 2xy \quad \frac{\partial^2 z}{\partial x \partial y} = 2x \quad \checkmark$

$\frac{\partial z}{\partial y} = x^2 + 3y^2 \quad \frac{\partial^2 z}{\partial y \partial x} = 2x \quad \checkmark$

$\frac{\partial^2 z}{\partial x^2} = 6x + 2y \quad \frac{\partial^2 z}{\partial y^2} = 6y$

定理: 若 $z = f(x, y)$ 它的不同顺序的

两个二阶偏导数 $f''_{xy}(x, y), f''_{yx}(x, y)$

在 (x_0, y_0) 处连续, 则 $f''_{xy}(x, y) = f''_{yx}(x, y)$

分析: $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f''_{xy}(x, y) = f''_{xy}(x_0, y_0)$

$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f''_{yx}(x, y) = f''_{yx}(x_0, y_0)$

$\exists \delta_0 > 0$, 使 $f''_{xy}(x, y), f''_{yx}(x, y)$ 在 $U(P_0, \delta_0)$ 内有定义。

构造 $\Delta W = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) - f(x_0, y_0 + \Delta y) + f(x_0, y_0)$

令 $\varphi(x) = f(x, y_0 + \Delta y) - f(x, y_0)$

$\Delta W = \varphi(x_0 + \Delta x) - \varphi(x_0)$ 由拉氏定理

$\Delta W = \varphi'(x_0 + \theta_1 \Delta x) \cdot \Delta x \quad (0 < \theta_1 < 1)$

... 波困难且不要求的操作

例: $z = f(2x - y) + g(\frac{y}{x})$ 且 f, g 具有二阶导数

求 $\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}$

解: $\frac{\partial z}{\partial x} = 2f'(2x - y) + g'(\frac{y}{x}) \cdot (-\frac{y}{x^2}) = 2f' + g' \cdot (-\frac{y}{x^2})$

$\frac{\partial^2 z}{\partial x^2} = 4f'' + g'' \cdot \frac{y^2}{x^4} + g' \cdot \frac{2y}{x^3} = 4f'' + \frac{y^2}{x^4} g'' + \frac{2y}{x^3} g'$

$\frac{\partial^2 z}{\partial x \partial y} = 2f'' \cdot (-1) + g''(\frac{y}{x}) \cdot (-\frac{y}{x^2}) + g' \cdot (-\frac{1}{x^2})$
 $= -2f'' - \frac{y}{x^3} g'' - \frac{1}{x^2} g'$





例: $u = \frac{1}{r} \quad r = \sqrt{x^2 + y^2 + z^2}$

求 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$

解: $\frac{\partial u}{\partial x} = -\frac{1}{r^2} \cdot \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2x$

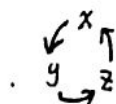
$$= -\frac{x}{r^3}$$

$$\frac{\partial^2 u}{\partial x^2} = -\frac{r^3 - 3xr^2 \cdot r'_x}{r^6}$$

$$= -\frac{r^3 - 3xr^2 \cdot \frac{x}{r}}{r^6}$$

$$= -\frac{1}{r^3} + \frac{3x^2}{r^5}$$

由轮换对称知



(大轮换: 同时 $x \rightarrow y, y \rightarrow z, z \rightarrow x$)

$$\frac{\partial^2 u}{\partial y^2} = -\frac{1}{r^3} + \frac{3y^2}{r^5} \quad \frac{\partial^2 u}{\partial z^2} = -\frac{1}{r^3} + \frac{3z^2}{r^5}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3(x^2+y^2+z^2) - 3r^2}{r^5} = 0$$

多元函数的全微分.

若 $y = f(x)$. 如果 $\Delta y = f(x_0 + \Delta x) - f(x_0)$ 可表示为

$\Delta y = A\Delta x + o(\Delta x) \quad (\Delta x \rightarrow 0)$ A 是与 Δx 无关的常数,

称 $y = f(x)$ 在 x_0 处可微. $A\Delta x$ 称为 $y = f(x)$

在 x_0 处微分. 记作 dy . 即 $dy = A\Delta x$.

$$o(\Delta x) = o(|\Delta x|) \quad (\Delta x \rightarrow 0) \quad \lim_{\Delta x \rightarrow 0} \frac{o(\Delta x)}{|\Delta x|} = \lim_{\Delta x \rightarrow 0} \frac{o(\Delta x)}{\pm \Delta x} = 0$$

$|\Delta x|$ 表示 $x_0 + \Delta x$ 与 x_0 之间的距离.

$$\Delta y = A\Delta x + o(|\Delta x|) \quad (\Delta x \rightarrow 0)$$

定义: 设 $z = f(x, y)$ 若 $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$

可表示为 $\Delta z = A\Delta x + B\Delta y + o(\rho) \quad (\rho \rightarrow 0)$

($\rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$) A, B 是与 $\Delta x, \Delta y$ 无关的常数.

称 $z = f(x, y)$ 在 (x_0, y_0) 处可微. 称线性主部

$A\Delta x + B\Delta y$ 为 $z = f(x, y)$ 在 (x_0, y_0) 处的全微分.

记作 dz . 即 $dz = A\Delta x + B\Delta y$.

定理: $z = f(x, y)$ 在 (x_0, y_0) 处可微. 则 $z = f(x, y)$ 在 (x_0, y_0) 连续.

$$\text{证: } \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \Delta z = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} (A\Delta x + B\Delta y + o(\rho)) = 0$$

故连续.

反之不成立

$$\text{注: } o(\rho) = o(\rho) \frac{\rho^2}{\rho^2} = o(\rho) \frac{\Delta x^2 + \Delta y^2}{\rho^2}$$

$$\therefore o(\rho) = \frac{o(\rho)}{\rho} \frac{\Delta x}{\rho} \cdot \Delta x + \frac{o(\rho)}{\rho} \frac{\Delta y}{\rho} \cdot \Delta y$$

$$\left(\left| \frac{\Delta x}{\rho} \right| = \frac{|\Delta x|}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = \frac{|\Delta x|}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \leq 1 \quad \left| \frac{\Delta y}{\rho} \right| \leq 1 \right)$$

$$\text{令 } \alpha = \frac{o(\rho)}{\rho} \frac{\Delta x}{\rho} \quad \beta = \frac{o(\rho)}{\rho} \frac{\Delta y}{\rho}$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \alpha = 0 \quad \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \beta = 0$$

$$\therefore o(\rho) = \alpha\Delta x + \beta\Delta y = o(\Delta x) + o(\Delta y) \quad \left(\begin{array}{l} \rho \rightarrow 0 \\ \text{或} \begin{array}{l} \Delta x \rightarrow 0 \\ \Delta y \rightarrow 0 \end{array} \end{array} \right)$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\alpha\Delta x}{\Delta x} = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \alpha = 0 \quad \therefore \alpha\Delta x = o(\Delta x)$$

$$\text{反之, } \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\alpha\Delta x + \beta\Delta y}{\rho} = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \left(\alpha \frac{\Delta x}{\rho} + \beta \frac{\Delta y}{\rho} \right) = 0$$

$$\text{故 } \Delta z = A\Delta x + B\Delta y + \alpha\Delta x + \beta\Delta y$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \alpha = 0 \quad \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \beta = 0$$





(可微的必要条件)

定理: 若 $z = f(x, y)$ 在 (x_0, y_0) 处可微.

则 $z = f(x, y)$ 在 (x_0, y_0) 处的两个偏导数

均存在. 且 $A = f'_x(x_0, y_0)$ $B = f'_y(x_0, y_0)$

反之不成立.

证: 由 $z = f(x, y)$ 在 (x_0, y_0) 处可微. 由可微定义.

$$\Delta z = A\Delta x + B\Delta y + \alpha\Delta x + \beta\Delta y \quad \lim_{\Delta x \rightarrow 0} \alpha = 0 \quad \lim_{\Delta y \rightarrow 0} \beta = 0$$

在(1)中令 $\Delta y = 0, \Delta x \neq 0$

$$\Delta x z = A\Delta x + \alpha\Delta x \quad \text{于是} \quad \lim_{\Delta x \rightarrow 0} \frac{\Delta x z}{\Delta x} = \lim_{\Delta x \rightarrow 0} (A + \alpha) = A$$

$$\therefore f'_x(x_0, y_0) = A. \quad \text{同理,} \quad f'_y(x_0, y_0) = B$$

反之不成立.

$$\text{例: } f(x, y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

在 $(0, 0)$ 处有 $f'_x(0, 0) = f'_y(0, 0) = 0$ 但在 $(0, 0)$ 处不连续.

(可微必连续. 不连续必不可微) 知在 $(0, 0)$ 处

不可微. 故反之不成立.

$A \Rightarrow B$ A 为 B 的充分. B 为 A 的必要

若 $z = f(x, y)$ 在 (x, y) 处可微则 $A = f'_x(x, y) = \frac{\partial z}{\partial x}$

$B = f'_y(x, y) = \frac{\partial z}{\partial y}$. 于是全微分 $dz = A\Delta x + B\Delta y$

$$dz = f'_x(x, y)\Delta x + f'_y(x, y)\Delta y$$

自变量的增量 = 自变量的微分. 有 $\Delta x = dx$ $\Delta y = dy$

$$\text{从而 } dz = f'_x(x, y)dx + f'_y(x, y)dy$$

$$= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

判断 $z = f(x, y)$ 在 (x, y) 处可微 $\Leftrightarrow$

$$\Delta z = A\Delta x + B\Delta y + o(\rho) \quad (\rho \rightarrow 0) \quad A, B \text{ 是只与 } \Delta x, \Delta y$$

无关的常数.

$$\Leftrightarrow \Delta z = f'_x(x, y)\Delta x + f'_y(x, y)\Delta y + o(\rho)$$

$$\Leftrightarrow \Delta z - f'_x(x, y)\Delta x - f'_y(x, y)\Delta y = o(\rho)$$

$$\Leftrightarrow \lim_{\rho \rightarrow 0} \frac{\Delta z - f'_x(x, y)\Delta x - f'_y(x, y)\Delta y}{\rho} = 0$$

$\Leftrightarrow$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) - f'_x(x_0, y_0)\Delta x - f'_y(x_0, y_0)\Delta y}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = 0$$

研究分片函数在孤立点 (x_0, y_0) 处.

定理 (可微的充分条件)

若 $z = f(x, y)$ 的两个偏导数 $f'_x(x, y)$ $f'_y(x, y)$ 在 (x_0, y_0) 处连续. 则 $z = f(x, y)$ 在 (x_0, y_0) 处可微. 反之不成立.

分析: 条件隐含

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f'_x(x, y) = f'_x(x_0, y_0)$$

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f'_y(x, y) = f'_y(x_0, y_0)$$

且 $\exists \delta > 0$ 当 $(x, y) \in U(\rho, \delta)$ 时

$f'_x(x, y)$ $f'_y(x, y)$ 有意义.

证明: 由 $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$

$$= f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y) + f(x_0, y_0 + \Delta y) - f(x_0, y_0)$$

$$= f'_x(x_0 + \theta_1 \Delta x, y_0 + \Delta y) \cdot \Delta x + f'_y(x_0, y_0 + \theta_2 \Delta y) \cdot \Delta y$$

$$(0 < \theta_1 < 1, 0 < \theta_2 < 1)$$

$$\left(\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y) = A \Leftrightarrow f(x, y) = A + \alpha, \quad \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} \alpha = 0 \right)$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} f'_x(x_0 + \theta_1 \Delta x, y_0 + \Delta y) = f'_x(x_0, y_0)$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} f'_y(x_0, y_0 + \theta_2 \Delta y) = f'_y(x_0, y_0)$$

$$\therefore \Delta z = (f'_x(x_0, y_0) + \alpha)\Delta x + (f'_y(x_0, y_0) + \beta)\Delta y$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \alpha = 0 \quad \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \beta = 0$$





$$\Delta z = f'_x(x_0, y_0) \Delta x + f'_y(x_0, y_0) \Delta y + o(\Delta x + \Delta y)$$

由可微的定义知, $z = f(x, y)$ 在 (x_0, y_0) 处可微. 反之不成立.

例: $u = ze^{\frac{x}{y}}$, 求 du

解: $\frac{\partial u}{\partial x} = z \cdot e^{\frac{x}{y}} \cdot \frac{1}{y}$

$$\frac{\partial u}{\partial y} = z e^{\frac{x}{y}} \cdot \left(-\frac{x}{y^2}\right)$$

$$\frac{\partial u}{\partial z} = e^{\frac{x}{y}}$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz$$

$$= \frac{z}{y} e^{\frac{x}{y}} dx - \frac{xz}{y^2} e^{\frac{x}{y}} dy + e^{\frac{x}{y}} dz$$

$$du|_{\substack{x=1 \\ y=1 \\ z=1}} = e dx - e dy + e dz$$

$$du|_{\substack{x=1, \Delta x=0.1 \\ y=1, \Delta y=0.2 \\ z=1, \Delta z=0.3}} = 0.1e - 0.2e + 0.3e$$

$$= 0.2e$$

微分在近似计算中的应用.

若 $z = f(x, y)$ 在 (x_0, y_0) 处可微, 即有

$$\Delta z = A \Delta x + B \Delta y + o(p)$$

$$= f'_x(x_0, y_0) \Delta x + f'_y(x_0, y_0) \Delta y + o(p)$$

$$= dz + o(p) \quad (p \rightarrow 0)$$

当 p 很小, $|o(p)|$ 更小, 于是, $\Delta z \approx dz$

因变量增量 Δz 约等于因变量的微分 dz . 于是

$$\Delta z \approx f'_x(x_0, y_0) \Delta x + f'_y(x_0, y_0) \Delta y$$

由于 $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$

$$f(x_0 + \Delta x, y_0 + \Delta y) \approx f(x_0, y_0) + f'_x(x_0, y_0) \Delta x + f'_y(x_0, y_0) \Delta y$$

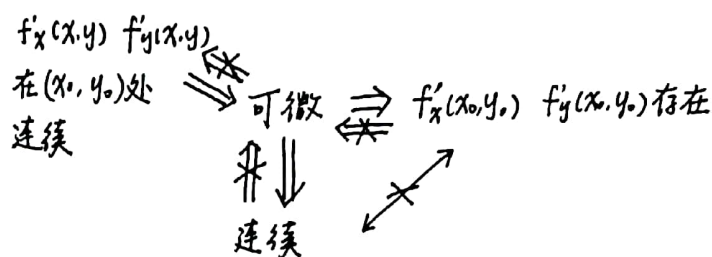
例: $z = f(x)$ 看成二元函数

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$= f'(x) dx + 0 \cdot dy$$

$$= f'(x) dx$$

以 $z = f(x, y)$ 在 (x_0, y_0) 处为例



多元复合函数的偏导数.

若 $z = f(u)$ 对 u 可导, $u = u(x, y)$ 对 x, y 偏导存在.

则 $z = f(u(x, y))$ 对 x, y 偏导数也存在且

$$\frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = f'(u) \frac{\partial u}{\partial x} \quad \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y}$$

若 $z = f(v)$ 对 v 可导, $v = v(x, y)$ 对 x, y 偏导存在.

则 $z = f(v(x, y))$ 对 x, y 的偏导数也存在且

$$\frac{\partial z}{\partial x} = \frac{dz}{dv} \cdot \frac{\partial v}{\partial x} = f'(v) \cdot \frac{\partial v}{\partial x} \quad \frac{\partial z}{\partial y} = \frac{dz}{dv} \cdot \frac{\partial v}{\partial y}$$

多元复合函数求偏导法则:

若 $z = f(u, v)$ $u = u(x, y)$ $v = v(x, y)$

$z = f(u(x, y), v(x, y))$ 且 z 在 (u, v) 处可微.

u, v 对 x, y 偏导数均存在, 则 $z = f(u(x, y), v(x, y))$ 对 x, y 的偏导数也存在.

$$\text{且 } \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$





证: 由 $z = f(u, v)$ 在 (u, v) 处可微

$$\Delta z = f(u + \Delta u, v + \Delta v) - f(u, v)$$

$$= \frac{\partial z}{\partial u} \Delta u + \frac{\partial z}{\partial v} \Delta v + \alpha \Delta u + \beta \Delta v \quad (1)$$

$$\lim_{\substack{\Delta u \rightarrow 0 \\ \Delta v \rightarrow 0}} \alpha = 0 \quad \lim_{\substack{\Delta u \rightarrow 0 \\ \Delta v \rightarrow 0}} \beta = 0$$

$$\Delta u = u(x + \Delta x, y + \Delta y) - u(x, y)$$

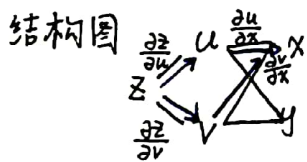
$$\Delta v = v(x + \Delta x, y + \Delta y) - v(x, y)$$

(1) 式中令 $\Delta y = 0$, $\Delta x \neq 0$ 有:

$$\Delta_x z = \frac{\partial z}{\partial u} \Delta u + \frac{\partial z}{\partial v} \Delta v + \alpha \Delta u + \beta \Delta v$$

$$\begin{aligned} \text{于是 } \lim_{\Delta x \rightarrow 0} \frac{\Delta_x z}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \left(\frac{\partial z}{\partial u} \cdot \frac{\Delta u}{\Delta x} + \frac{\partial z}{\partial v} \cdot \frac{\Delta v}{\Delta x} + \alpha \frac{\Delta u}{\Delta x} + \beta \frac{\Delta v}{\Delta x} \right) \\ &= \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} + 0 \cdot \frac{\partial u}{\partial x} + 0 \cdot \frac{\partial v}{\partial x} \\ &= \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \\ &= \frac{\partial z}{\partial x} \end{aligned}$$

同理, $\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$



$$\frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x}$$

例: $z = f(u, v, w)$ 可微.

$$u = u(x, y, t) \quad v = v(x, y, t) \quad w = w(x, y, t)$$

偏导数均存在, $z = f(u(x, y, t), v(x, y, t), w(x, y, t))$

偏导数也存在且

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} + \frac{\partial z}{\partial w} \cdot \frac{\partial w}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} + \frac{\partial z}{\partial w} \cdot \frac{\partial w}{\partial y}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial t} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial t} + \frac{\partial z}{\partial w} \cdot \frac{\partial w}{\partial t}$$

例: $z = f(x, u, v)$ 可微, $u = u(x, y)$ $v = v(x, y)$

偏导数均存在 则 $z = f(x, u(x, y), v(x, y))$

偏导数也存在.

注: $f(x, u, v)$ 是三元函数

$$f(x, u, v) = x + u^2 v + v$$

$$f'_x(x, u, v) = 1$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

这种写法并不正确.

正确写法:

$$\frac{\partial z}{\partial x} = f'_x(x, u, v) + f'_u(x, u, v) \frac{\partial u}{\partial x} + f'_v(x, u, v) \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = f'_u(x, u, v) \cdot \frac{\partial u}{\partial y} + f'_v(x, u, v) \frac{\partial v}{\partial y}$$

$$= \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}$$

例: $z = f(u, v, w)$ 可微.

$$u = u(x) \quad v = v(x) \quad w = w(x) \text{ 均可导}$$

则 $z = f(u(x), v(x), w(x))$ 对 x 可导.

$$\frac{dz}{dx} = \frac{\partial z}{\partial u} \cdot \frac{du}{dx} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dx} + \frac{\partial z}{\partial w} \cdot \frac{dw}{dx}$$

称为全导数.

例: $z = (x^2 + y^2)^{xy}$ 求 $\frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y}$

解: 设 $u = x^2 + y^2$ $v = xy$ $z = u^v$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= \frac{\partial z}{\partial u} \cdot v \cdot u^{v-1} \cdot (2x) + u^v \cdot \ln u \cdot y$$

$$= u^v \left(\frac{v}{u} \cdot 2x + y \cdot \ln u \right)$$

$$= (x^2 + y^2)^{xy} \left[\frac{2x^2 y}{x^2 + y^2} + y \cdot \ln(x^2 + y^2) \right]$$

同理. 由轮换

$$\frac{\partial z}{\partial y} = (x^2 + y^2)^{xy} x \left(\frac{2y^2}{x^2 + y^2} + \ln(x^2 + y^2) \right)$$





例: $z = f(x^2 - y^2, xy)$ 且 f 具有连续的 n 阶偏导数. 求 $\frac{\partial^2 z}{\partial x \partial y}$

解: 设 $u = x^2 - y^2$ $v = xy$ $z = f(u, v)$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$= f'_u(u, v) \cdot 2x + f'_v(u, v) \cdot y$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= 2x \cdot [f''_{uv}(u, v) \cdot (-2y) + f''_{vu}(u, v) \cdot x] \\ &\quad + [f'_{vu}(u, v) \cdot (-2y) + f'_{vv}(u, v) \cdot x] \cdot y \\ &\quad + f''_v(u, v) \cdot 1 \end{aligned}$$

$$= -4xy f''_{uv}(u, v) + 2(x^2 - y^2) f''_{vu}(u, v) + f''_v(u, v)$$

(注:)

解: $z = f(x^2 - y^2, xy)$ $f'_i = f'_i(x^2 - y^2, xy)$

$$\frac{\partial z}{\partial x} = f'_1 \cdot 2x + f'_2 \cdot y$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= 2x [f''_{12}(x^2 - y^2, xy) + f''_{21}(x^2 - y^2, xy) \cdot x] \\ &\quad + [f'_{21}(x^2 - y^2, xy) \cdot (-2y) + f'_{22}(x^2 - y^2, xy) \cdot x] \cdot y + f''_2(x^2 - y^2, xy) \end{aligned}$$

例: $u = f(x+y+z, x^2+y^2+z^2)$ f 具有连续的 n 阶偏导数. 求 $\frac{\partial^2 u}{\partial x^2}$

解: $\frac{\partial u}{\partial x} = f'_1 + f'_2 \cdot 2x$

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= f''_{11} + f''_{12} \cdot 2x + 2x [f''_{21} + f''_{22} \cdot 2x] + 2f''_2 \\ &= f''_{11} + 4x f''_{12} + 4x^2 f''_{22} + 2f''_2 \end{aligned}$$

例: $z = x^3 f(xy, \frac{y}{x})$ 且 f 具有连续的 n 阶偏导数. 求 $\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}$

解: $\frac{\partial z}{\partial x} = x^3 [f'_1 \cdot x + f'_2 \cdot \frac{y}{x}] = x^4 f'_1 + x^2 f'_2$

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= x^4 (f''_{11} \cdot x + f''_{12} \cdot \frac{y}{x}) + x^2 (f''_{21} \cdot x + f''_{22} \cdot \frac{y}{x}) \\ &= x^5 f''_{11} + 2x^3 f''_{12} + x f''_{22} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial^2 z}{\partial y \partial x} \\ &= 4x^3 f'_1 + x^2 [f''_{11} y + f''_{12} (\frac{y}{x})] \\ &\quad + 2x f'_2 + x^2 [f''_{21} y + f''_{22} (\frac{y}{x})] \end{aligned}$$

对全微分进一步讨论.

1. 设 u, v 为多元函数. 若 u, v 均可微. 则

$$d(u \pm v) = du \pm dv$$

$$d(uv) = vdu + u dv$$

$$d(\frac{u}{v}) = \frac{vdu - udv}{v^2}$$

$$d(\frac{1}{v}) = -\frac{dv}{v^2}$$

2. 若 $z = z(x, y)$ 可微. 当 x, y 为自变量时.

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

当 $x = x(s, t)$ $y = y(s, t)$ 时可以证明

$z = z(x(s, t), y(s, t))$ 在 (s, t) 处可微.

$$\begin{aligned} dz &= \frac{\partial z}{\partial s} ds + \frac{\partial z}{\partial t} dt \\ &= (\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}) ds + (\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}) dt \\ &= \frac{\partial z}{\partial x} (\frac{\partial x}{\partial s} ds + \frac{\partial x}{\partial t} dt) + \frac{\partial z}{\partial y} (\frac{\partial y}{\partial s} ds + \frac{\partial y}{\partial t} dt) \\ &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \end{aligned}$$

总结: 若 $z = z(x, y)$ 可微, 不论 x, y 为自变量或中间变量, 都有 $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$. 称为全微分的一阶形式不变性.

例: 若 $z = z(u, v)$ 可微. 若 $dz = g(u, v) du + h(u, v) dv$

$$\text{则 } \frac{\partial z}{\partial u} = g(u, v) \quad \frac{\partial z}{\partial v} = h(u, v)$$

例: $z = f(\cos \theta, r \sin \theta)$ 且 f 有连续的 n 阶偏导数. 求 $dz, \frac{\partial z}{\partial r}, \frac{\partial z}{\partial \theta}$





$$dz = d f(r \cos \theta, r \sin \theta)$$

$$= f'_1 d(r \cos \theta) + f'_2 d(r \sin \theta)$$

$$= f'_1 (\cos \theta dr - r \sin \theta d\theta) + f'_2 (\sin \theta dr + r \cos \theta d\theta)$$

$$= f'_1 [\cos \theta dr - r \sin \theta d\theta] + f'_2 [\sin \theta dr + r \cos \theta d\theta]$$

$$= f'_1 \cos \theta dr - f'_1 r \sin \theta d\theta + f'_2 \sin \theta dr + f'_2 r \cos \theta d\theta$$

$$= (f'_1 \cos \theta + f'_2 \sin \theta) dr + (f'_2 r \cos \theta - f'_1 r \sin \theta) d\theta$$

$$\therefore \frac{\partial z}{\partial r} = f'_1 \cos \theta + f'_2 \sin \theta$$

$$\frac{\partial z}{\partial \theta} = -r f'_1 \sin \theta + r f'_2 \cos \theta$$

解法二: $\frac{\partial z}{\partial r} = f'_1 \cos \theta + f'_2 \sin \theta$

$$\frac{\partial z}{\partial \theta} = f'_1 r (-\sin \theta) + f'_2 r \cos \theta$$

$$= -r f'_1 \sin \theta + r f'_2 \cos \theta$$

$$dz = \frac{\partial z}{\partial r} dr + \frac{\partial z}{\partial \theta} d\theta$$

$$= (f'_1 \cos \theta + f'_2 \sin \theta) dr + (-r f'_1 \sin \theta + r f'_2 \cos \theta) d\theta$$

方程确定多元函数的偏导数

三元方程 $F(x, y, z) = 0$, 设 $D \subset \mathbb{R}^2, D \neq \emptyset, C \subset \mathbb{R}, C \neq \emptyset$

$\forall P_0(x_0, y_0) \in D$, 代入 (1) 有 $F(x_0, y_0, z) = 0$ 如果

方程有唯一解 $z = z_0 \in C$ 即 $F(x_0, y_0, z_0) = 0$

由二元函数的定义, 确定 D 上的一个二元函数, 记为

$z = z(x, y) (x, y) \in D$ 有 $F(x, y, z(x, y)) \equiv 0$ 称为

$z = z(x, y)$ 是由方程 $F(x, y, z) = 0$ 确定的函数.

如果从方程中能将 z 解出来用 x, y 表示, 称为显函数.

如果 z 没有从方程中解出用 x, y 表示, 称 $z = z(x, y)$

为隐函数.

若 $F(x, y, z) = 0$ 确定 $z = z(x, y)$, 如何

$$\text{求 } \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$$

分析: 由 $F(x, y, z) = F(x, y, z(x, y)) \equiv 0$

要求 $F(x, y, z)$ 具有连续的一阶偏导数.

方程两边对 x 偏导.

$$F'_x \cdot 1 + F'_z \cdot \frac{\partial z}{\partial x} = 0 \therefore \frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z}$$

$$\text{即 } \frac{\partial z}{\partial x} = -\frac{F'_x(x, y, z)}{F'_z(x, y, z)} \quad \text{同理}$$

$$\frac{\partial z}{\partial y} = -\frac{F'_y(x, y, z)}{F'_z(x, y, z)}$$

总结: 若 $F(x, y, z)$ 在 $U(P_0)$ ($P_0(x_0, y_0, z_0)$

$F(x_0, y_0, z_0) = 0$) 内具有连续的一阶偏导数.

且 $F'_z(x, y, z) \neq 0$ 则 $\frac{\partial z}{\partial x} = -\frac{F'_x(x, y, z)}{F'_z(x, y, z)}$

$$\frac{\partial z}{\partial y} = -\frac{F'_y(x, y, z)}{F'_z(x, y, z)} \quad \text{注: 偏导不为0者视为“z”}$$

若 $F(x, y, z) = 0$ 确定 $x = x(y, z)$ 则

$$\frac{\partial x}{\partial y} = -\frac{F'_y(x, y, z)}{F'_x(x, y, z)} \quad \frac{\partial x}{\partial z} = -\frac{F'_z(x, y, z)}{F'_x(x, y, z)}$$

若 $F(x, x_1, \dots, x_n, y) = 0$ 确定了

$y = y(x_1, x_2, \dots, x_n)$ 的函数.

$$\frac{\partial y}{\partial x_i} = -\frac{F'_{x_i}(x_1, x_2, \dots, x_n, y)}{F'_y(x_1, x_2, \dots, x_n, y)}$$

若二元方程 $F(x, y) = 0$ 确定 $y = y(x)$

$$\text{求 } \frac{dy}{dx} = -\frac{F'_x(x, y)}{F'_y(x, y)}$$

如果确定 $x = x(y)$

$$\frac{dx}{dy} = -\frac{F'_y(x, y)}{F'_x(x, y)}$$





例: $z^3 - 3xyz = a^3$ (a 为常)

求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

解法一: $z^3 - 3xyz - a^3 = 0$

设 $F(x, y, z) = z^3 - 3xyz - a^3$

$$\frac{\partial z}{\partial x} = - \frac{F'_x(x, y, z)}{F'_z(x, y, z)} = - \frac{-3yz}{3z^2 - 3xy} = \frac{yz}{z^2 - xy}$$

同理, 轮换得 $\frac{\partial z}{\partial y} = \frac{xz}{z^2 - xy}$

解法二: 由 $z = z(x, y)$ 方程两边对 x 偏导.

$$3z^2 \frac{\partial z}{\partial x} - 3y(z + x \frac{\partial z}{\partial x}) = 0$$

$$(z^2 - xy) \frac{\partial z}{\partial x} = yz \quad \therefore \frac{\partial z}{\partial x} = \frac{yz}{z^2 - xy}$$

同理 $\frac{\partial z}{\partial y} = \frac{xz}{z^2 - xy}$.

解法三: 由 $z = z(x, y)$ 由

$$d(z^3 - 3xyz) = d(a^3)$$

$$d(z^3) - 3d(xyz) = 0$$

$$3z^2 dz - 3(yz dx + xz dy + xy dz) = 0$$

全微分-阶形式不变性:

$$dF(u, v, w) = F'_u du + F'_v dv + F'_w dw$$

$$d(xyz) = yz dx + xz dy + xy dz$$

$$(z^2 - xy) dz = yz dx + xz dy$$

$$dz = \frac{yz}{z^2 - xy} dx + \frac{xz}{z^2 - xy} dy$$

$$\therefore \frac{\partial z}{\partial x} = \frac{yz}{z^2 - xy} \quad \frac{\partial z}{\partial y} = \frac{xz}{z^2 - xy}$$

2个方程
4个变量

↓
确定2个元函数

例: $\begin{cases} xu - yv = 0 \\ yu + xv = 1 \end{cases}$ 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$

解: 由条件知 $u = u(x, y)$ $v = v(x, y)$

方程组两边对 x 偏导

$$\begin{cases} u + x \frac{\partial u}{\partial x} - y \frac{\partial v}{\partial x} = 0 \\ y \frac{\partial u}{\partial x} + v + x \frac{\partial v}{\partial x} = 0 \end{cases} \quad \begin{cases} x \cdot \frac{\partial u}{\partial x} - y \cdot \frac{\partial v}{\partial x} = -u \\ y \frac{\partial u}{\partial x} + x \frac{\partial v}{\partial x} = -v \end{cases}$$

例: $F(x-y, y-z, z-x) = 0$

F 具有连续的一阶偏导数求 $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$.

解: 由条件知 $z = z(x, y)$. 两边微分得

$$dF(x-y, y-z, z-x) = d0 = 0$$

$$F'_1 d(x-y) + F'_2 d(y-z) + F'_3 d(z-x) = 0$$

$$F'_1 (dx - dy) + F'_2 (dy - dz) + F'_3 (dz - dx) = 0$$

$$(F'_1 - F'_3) dx + (F'_2 - F'_1) dy + (F'_3 - F'_2) dz = 0$$

$$dz = \frac{F'_1 - F'_3}{F'_2 - F'_3} dx + \frac{F'_2 - F'_1}{F'_2 - F'_3} dy$$

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \frac{F'_2 - F'_3}{F'_2 - F'_3} = 1$$

解法二: $F(x-y, y-z, z-x) = 0$

$$\frac{\partial z}{\partial x} = - \frac{F'_x}{F'_z} = - \frac{F'_1 - F'_3}{-F'_2 + F'_3} = \frac{F'_1 - F'_3}{F'_2 - F'_3}$$

$$\frac{\partial z}{\partial y} = - \frac{F'_y}{F'_z} = - \frac{-F'_1 + F'_2}{-F'_2 + F'_3} = \frac{F'_2 - F'_1}{F'_2 - F'_3}$$

$$\therefore \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 1$$

方程组确定多元函数组的求偏导问题.

例: 4个方程, 9个变量, 确定 (4) 个 (5) 元多元函数





由克拉默法则知:

$$\frac{\partial u}{\partial x} = \frac{\begin{vmatrix} -u & -y \\ -v & x \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = -\frac{ux+vy}{x^2+y^2}$$

$$\frac{\partial v}{\partial x} = \frac{\begin{vmatrix} x & -u \\ y & -v \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = \frac{uy-vx}{x^2+y^2}$$

解法: $\begin{cases} xu-yv=0 \\ yu+xv=1 \end{cases}$ 两边微分

得: $\begin{cases} d(xu)-d(yv)=0 \\ d(yu)+d(xv)=0 \end{cases}$

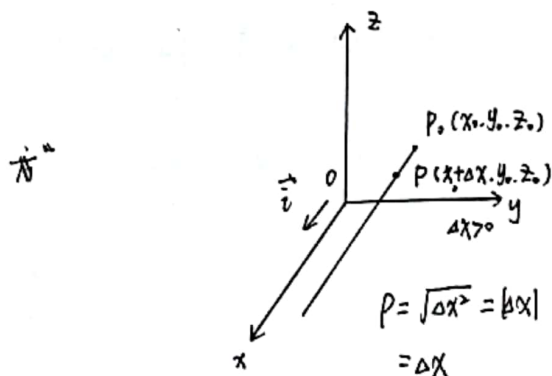
$$\begin{cases} udx+xdu-vdy-ydv=0 \\ udy+ydu+vdx+xdv=0 \end{cases}$$

$$\begin{cases} xdu-ydv=-udx+vdy \\ ydu+xdv=-vdx-udy \end{cases}$$

由克拉默法则知:

$$du = \frac{\begin{vmatrix} -udx+vdv & -y \\ -vdx-udy & x \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$dv = \frac{\begin{vmatrix} x & -udx+vdv \\ y & -vdx-udy \end{vmatrix}}{\begin{vmatrix} x & -y \\ y & x \end{vmatrix}} = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$$



方向导数及梯度

以 $z=f(x,y)$ 在 (x_0, y_0) 处为例. 若 $f'_x(x_0, y_0)$ 存在

$$\begin{aligned} &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0+\Delta x, y_0) - f(x_0, y_0)}{\Delta x} = \lim_{x \rightarrow x_0} \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \\ &= \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0+\Delta y) - f(x_0, y_0)}{\Delta y} \\ &= \lim_{y \rightarrow y_0} \frac{f(x_0, y) - f(x_0, y_0)}{y - y_0} \end{aligned}$$

方向导数定义: (以三元函数为例)



设 $P_0(x_0, y_0, z_0) \in R^3$. $\exists \delta > 0$. $u=f(x, y, z)$

在 $U(P_0, \delta)$ 内有定义. l 是从 P_0 发出的一条射线.

$\forall P(x, y, z) \in l \cap U(P_0, \delta)$ ($P \neq P_0$) 设 $x-x_0=\Delta x$

$y-y_0=\Delta y$. $z-z_0=\Delta z$. $\therefore P(x_0+\Delta x, y_0+\Delta y, z_0+\Delta z)$

若 $\lim_{\rho \rightarrow 0} \frac{u(P)-u(P_0)}{\rho}$ (存在) $(|P-P_0| = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \triangleq \rho)$

$$= \lim_{\rho \rightarrow 0} \frac{u(x_0+\Delta x, y_0+\Delta y, z_0+\Delta z) - u(x_0, y_0, z_0)}{\rho}$$

$$= \lim_{\rho \rightarrow 0} \frac{u(x_0+\Delta x, y_0+\Delta y, z_0+\Delta z) - u(x_0, y_0, z_0)}{\rho}$$

$$= \lim_{\rho \rightarrow 0} \frac{\Delta u}{\rho} \text{ (存在)} \triangleq \frac{\partial u}{\partial l} \Big|_{P=P_0} = u_l(P_0)$$

称为 $u(x, y, z)$ 在 $P=P_0$ 处沿着方向 l 的方向导数.

$$\text{若 } \frac{\partial u}{\partial x} \Big|_{P=P_0} \text{ 存在 } \lim_{\Delta x \rightarrow 0} \frac{u(x_0+\Delta x, y_0, z_0) - u(x_0, y_0, z_0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0^+} \frac{u(x_0+\Delta x, y_0, z_0) - u(x_0, y_0, z_0)}{\Delta x}$$

$$= \frac{\partial u}{\partial x} \Big|_{P=P_0} = \frac{\partial u}{\partial x} \Big|_{P=P_0}$$





同理, $\frac{\partial u}{\partial (-i)}|_{P=P_0} = -\frac{\partial u}{\partial x}|_{P=P_0}$.

若 $\frac{\partial u}{\partial y}|_{P=P_0}$ 存在, 则 $\frac{\partial u}{\partial j}|_{P=P_0} = \frac{\partial u}{\partial y}|_{P=P_0}$.

且 $\frac{\partial u}{\partial (-j)}|_{P=P_0} = -\frac{\partial u}{\partial y}|_{P=P_0}$.

若 $\frac{\partial u}{\partial z}|_{P=P_0}$ 存在, 则 $\frac{\partial u}{\partial k}|_{P=P_0} = \frac{\partial u}{\partial z}|_{P=P_0}$.

且 $\frac{\partial u}{\partial (-k)}|_{P=P_0} = -\frac{\partial u}{\partial z}|_{P=P_0}$.

方向导数存在的充分条件.

定理: 若 $u = u(x, y, z)$ 在 $P_0(x_0, y_0, z_0)$ 处可微.

则 u 在 P_0 点沿任意方向 $\vec{l}$ 的方向导数均存在.

$$\frac{\partial u}{\partial \vec{l}}|_{P_0} = \frac{\partial u}{\partial x}|_{P_0} \cos \alpha + \frac{\partial u}{\partial y}|_{P_0} \cos \beta + \frac{\partial u}{\partial z}|_{P_0} \cos \gamma$$

其中 $\vec{l} = \{\cos \alpha, \cos \beta, \cos \gamma\}$

证明: 由条件 u 在 P_0 处可微定义知 $\vec{l}$

$$\Delta u = u(x_0 + \Delta x, y_0 + \Delta y, z_0 + \Delta z) - u(x_0, y_0, z_0)$$

$$= \frac{\partial u}{\partial x} \Delta x + \frac{\partial u}{\partial y} \Delta y + \frac{\partial u}{\partial z} \Delta z + o(\rho)$$

$\vec{P_0 P} \parallel \vec{l}$ 且方向一致 $\Rightarrow \vec{P_0 P} = \vec{l} \cdot \rho$

$$\vec{P_0 P} = \{\Delta x, \Delta y, \Delta z\} \quad \vec{P_0 P} = \left\{ \frac{\Delta x}{\rho}, \frac{\Delta y}{\rho}, \frac{\Delta z}{\rho} \right\} \cdot \rho$$

$$= \{\cos \alpha, \cos \beta, \cos \gamma\} \cdot \rho$$

$$\frac{\Delta x}{\rho} = \cos \alpha \quad \frac{\Delta y}{\rho} = \cos \beta \quad \frac{\Delta z}{\rho} = \cos \gamma$$

$$\begin{aligned} \text{于是 } \lim_{\rho \rightarrow 0} \frac{\Delta u}{\rho} &= \lim_{\rho \rightarrow 0} \left[\frac{\partial u}{\partial x}|_{P_0} \frac{\Delta x}{\rho} + \frac{\partial u}{\partial y}|_{P_0} \frac{\Delta y}{\rho} + \frac{\partial u}{\partial z}|_{P_0} \frac{\Delta z}{\rho} + \frac{o(\rho)}{\rho} \right] \\ &= \frac{\partial u}{\partial x}|_{P_0} \cos \alpha + \frac{\partial u}{\partial y}|_{P_0} \cos \beta + \frac{\partial u}{\partial z}|_{P_0} \cos \gamma \\ &= \frac{\partial u}{\partial \vec{l}}|_{P_0} \end{aligned}$$

反之不成立.

如果求分片函数在孤立点的方向导数, 一般用方向导数的定义.

定义: 若 $u = u(x, y, z)$ 在 P_0 点的三个偏导数均存在, 记 $\left\{ \frac{\partial u}{\partial x}|_{P_0}, \frac{\partial u}{\partial y}|_{P_0}, \frac{\partial u}{\partial z}|_{P_0} \right\} = \text{grad } u|_{P=P_0}$ 称为 u 在 $P=P_0$ 处的梯度 (该梯度为一向量)

若 $u = u(x, y, z)$ 在区域 G 上三个偏导数均存在, 记 $\left\{ \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right\} = \text{grad } u$ 称为 u 的梯度.

$$\text{且 } \text{grad } u|_{P=P_0} = \left\{ \frac{\partial u}{\partial x}|_{P_0}, \frac{\partial u}{\partial y}|_{P_0}, \frac{\partial u}{\partial z}|_{P_0} \right\} = \left\{ \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right\}|_{P=P_0} = \text{grad } u(P_0)$$

若 $\text{grad } u, \text{grad } v$ 均存在, 则对于任何常数 α, β 都有 $\text{grad}(\alpha u + \beta v) = \alpha \text{grad } u + \beta \text{grad } v$

如果 $u(x, y, z)$ 在 P_0 处可微.

$$\frac{\partial u}{\partial \vec{l}}|_{P_0} = \text{grad } u(P_0) \cdot \vec{l}^0 \quad \text{几何意义为:}$$

若 $\frac{\partial u}{\partial \vec{l}}|_{P_0}$ 存在, 则 $\frac{\partial u}{\partial \vec{l}}|_{P_0}$ 表示 $\text{grad } u(P_0)$ 在 $\vec{l}$ 上的投影. 实际意义为: 若 $\frac{\partial u}{\partial \vec{l}}|_{P_0}$ 存在, 指 u 在 P_0 沿方向 $\vec{l}$ 的变化率.

若 $u = u(x, y, z)$ 在 P_0 点可微

$$\begin{aligned} \frac{\partial u}{\partial \vec{l}}|_{P_0} &= \text{grad } u(P_0) \cdot \vec{l}^0 \\ &= |\text{grad } u(P_0)| \cdot |\vec{l}^0| \cdot \cos \theta \\ &= |\text{grad } u(P_0)| \cdot \cos \theta \end{aligned}$$

其中 θ 指的是 $\vec{l}$ 与 $\text{grad } u(P_0)$ 的夹角, $0 \leq \theta \leq \pi$

$$|\text{grad } u(P_0)| = \sqrt{\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2}$$





$|\text{grad} u(p_0)| \geq 0$ (常数) 当 $\theta = 0^+$ 时, 即 $\vec{e}$ 与 $\text{grad} u(p_0)$ 方向一致, 即 $u(x, y, z)$ 在 P_0 处沿 $\text{grad} u(p_0)$ 方向时方向导数最大.

$\max \left\{ \frac{\partial u}{\partial \vec{e}} \Big|_{P_0} \right\} = |\text{grad} u(p_0)|$ 当 $\theta = \pi$ 时.

$\vec{e}$ 与 $\text{grad} u(p_0)$ 方向相反, 即 $u(x, y, z)$ 在 P_0 处沿 $-\text{grad} u(p_0)$ 方向时方向导数最小.

$\min \left\{ \frac{\partial u}{\partial \vec{e}} \Big|_{P_0} \right\} = -|\text{grad} u(p_0)|$ 当 $\theta = \frac{\pi}{2}$ 时.

即 $u(x, y, z)$ 在 P_0 处沿与 $\text{grad} u(p_0)$ 垂直方向的方向导数为 0.

例: 求 $u = xy^2 + z^3 - xyz$ 在 $P_0(1, 1, 1)$ 沿哪个方向方向导数最大, 最小, 为 0, 并求值.

解: $\frac{\partial u}{\partial x} = y^2 - yz \quad \frac{\partial u}{\partial x} \Big|_{P_0} = 0$

$\frac{\partial u}{\partial y} = 2xy - xz \quad \frac{\partial u}{\partial y} \Big|_{P_0} = 1$

$\frac{\partial u}{\partial z} = 3z^2 - xy \quad \frac{\partial u}{\partial z} \Big|_{P_0} = 2$

从而, $\text{grad} u(p_0) = \{0, 1, 2\}$

$|\text{grad} u(p_0)| = \sqrt{5}$, 沿 $\{0, 1, 2\}$ 方向的方向导数

$\max = \sqrt{5}$ 沿 $-\{0, 1, 2\}$ 方向的方向导数最小

$\min = -\sqrt{5}$

设沿 $\{a, b, c\}$ 方向的方向导数为 0.

$\therefore \{a, b, c\} \cdot \{0, 1, 2\} = 0 \quad \therefore b + 2c = 0$

$\therefore b = -2c \quad \therefore$ 沿 $\{a, -2c, c\}$ (a, c 不同时为零)

方向导数为 0

例: $z = xy - y^2$ 在 $P_0(1, 1)$ 处沿

$\vec{e} = \{1, 2\}$ 方向的方向导数.

解: $\frac{\partial z}{\partial x} = y \quad \frac{\partial z}{\partial x} \Big|_{P_0} = 1 \quad \frac{\partial z}{\partial y} = x - 2y \quad \frac{\partial z}{\partial y} \Big|_{P_0} = -1$

$|\text{grad} z| = \sqrt{2} \quad \vec{e} = \left\{ \frac{1}{\sqrt{2}}, \frac{2}{\sqrt{2}} \right\}$

于是 $\frac{\partial z}{\partial \vec{e}} \Big|_{P_0} = 1 \times \frac{1}{\sqrt{2}} + (-1) \times \frac{2}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$

多元函数的极值及其应用

多元函数的泰勒公式

定理: 设 $z = f(x, y)$ 在区域 G 上偏导数存在.

- (1) 若 $f'_x(x, y) \equiv 0 \quad (x, y) \in G$, 则 z 仅是 y 的函数
- (2) 若 $f'_y(x, y) \equiv 0 \quad (x, y) \in G$, 则 z 仅是 x 的函数
- (3) 若 $f'_x(x, y) \equiv 0 \quad f'_y(x, y) \equiv 0$, 则 z 是常值函数.

以 $z = f(x, y)$ 为例. 设 $P_0(x_0, y_0) \in R^2$

定义: 若 $\exists \delta_0 > 0$, 当 $P(x, y) \in U(P_0, \delta_0)$ 时

都有 $f(x, y) \leq f(x_0, y_0)$ 称 $f(x_0, y_0)$ 为极大值

称 (x_0, y_0) 为极大值点. 都有 $f(x, y) \geq f(x_0, y_0)$

称 $f(x_0, y_0)$ 为极小值, 称 (x_0, y_0) 为极小值点.

极大值与极小值统称为极值. 极大值点, 极小值点

统称为极值点 (极值点一定包含在区域的内部)

定理: 若 $z = f(x, y)$ 在 $P_0(x_0, y_0)$ 处取到极值

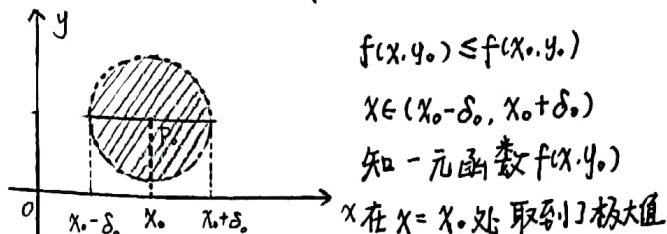
且 $f'_x(x_0, y_0), f'_y(x_0, y_0)$ 均存在, 则 $f'_x(x_0, y_0) = 0$,

$f'_y(x_0, y_0) = 0$ 反之不成立 (取到极值的必要条件)





证明: 由条件知 $z=f(x,y)$ 在 $P_0(x_0, y_0)$ 处
取到极值, 不妨设取到极大值, 由定义
 $\exists \delta_0 > 0$, 当 $P(x,y) \in U(P_0, \delta_0)$ 时, $f(x,y) \leq f(x_0, y_0)$



且 $\frac{df(x,y)}{dx} \Big|_{x=x_0} = f'_x(x_0, y_0)$ 存在, 由费马定理

则 $f'_x(x_0, y_0) = 0$, 同理 $f'_y(x_0, y_0) = 0$

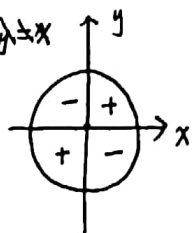
反之不成立

例: $f(x,y) = xy$ ~~$f'_x = y, f'_y = x$~~

$f'_x(x,y) = y$ $f'_x(0,0) = 0$

$f'_y(x,y) = x$ $f'_y(0,0) = 0$

但 $f(0,0) = 0$ 不是极值.



定义: 若 $f'_x(x_0, y_0) = 0$ $f'_y(x_0, y_0) = 0$ 则称 $P_0(x_0, y_0)$
为驻点或稳定点.

总结: $z=f(x,y)$ 的极值点一定包含在区域内部的
驻点或内部偏导数至少有一个不存在点之中.

例: $z = \sqrt{x^2 + y^2}$ $z(0,0) = 0$, $z(x,y) \geq z(0,0)$

$x, y \in \mathbb{R}^2$ 知 $z(0,0) = 0$ 为极小值, 但是

$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}$ $\frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$ z 在 $(0,0)$ 处的
偏导数不存在.

对于驻点 $P_0(x_0, y_0)$ 有下面的定理:

定理 (取到极值的充分条件)

若在 (x_0, y_0) 处 $f'_x(x_0, y_0) = 0$ $f'_y(x_0, y_0) = 0$

设 $z=f(x,y)$ 在 $U(P_0, \delta)$ ($\delta > 0$, 常) 内具有连续
的二阶偏导数. 设 $f''_{xx}(x_0, y_0) = A$ $f''_{xy}(x_0, y_0) = B$
 $f''_{yy}(x_0, y_0) = C$

(1) $B^2 - AC < 0$ 则 $f(x_0, y_0)$ 为极值, 此时 A, C 同号

当 $A > 0$, 则 $f(x_0, y_0)$ 为极小值

当 $A < 0$, 则 $f(x_0, y_0)$ 为极大值

(2) $B^2 - AC > 0$ 则 $f(x_0, y_0)$ 不是极值

(3) $B^2 - AC = 0$ 本方法失效.

$P(x_0 + h, y_0 + k) \in U(P_0, \delta)$

$f(x_0 + h, y_0 + k) = f(x_0, y_0) + \dots$ (-阶偏导为0)

$f(x_0 + h, y_0 + k) - f(x_0, y_0) =$ 3个三阶偏导数.

↓

差的符号由 $Ah^2 + 2Bhk + Ck^2$ 确定
(看成 h 的一元二次函数)

(1) $B^2 - AC < 0$

$\Delta = 4k^2(B^2 - AC) < 0$ 抛物线与 x 轴无交点

$A > 0$, $Ah^2 + 2Bhk + Ck^2 > 0$, 则

$f(x_0 + h, y_0 + k) > f(x_0, y_0)$, $f(x_0, y_0)$ 为极小值.

$A < 0$ $f(x_0 + h, y_0 + k) < f(x_0, y_0)$, $f(x_0, y_0)$ 为极大值

(2) $B^2 - AC > 0$ 非极值

(3) $B^2 - AC = 0$ 可取极值, 可不取, 方法失效

例: 求 $f(x,y) = y^3 - x^2 + 6x - 12y + 5$ 的极值

解: $\begin{cases} f'_x = -2x + 6 = 0 \\ f'_y = 3y^2 - 12 = 0 \end{cases} \begin{cases} x = 3 \\ y = \pm 2 \end{cases}$ 显然, 在内部
且无偏导不存在点

$f''_{xx} = -2$ $f''_{xy} = 0$ $f''_{yy} = 6y$

在 $(3, 2)$ 点 $B = 0$ $A = -2$ $C = 12$ $B^2 - AC > 0$ 非极值

在 $(3, -2)$ 点 $B = 0$ $A = -2$ $C = -12$ $B^2 - AC < 0$

$A = -2 < 0$ $f_{\max}(x,y) = f(3, -2) = 30$

19.





注: (3,2) 为唯一-的极值, 但并非最值

若 $u=f(p)$ 在有界闭区域 G 上连续 ($G \subset \mathbb{R}^n$)

则 $u=f(p)$ 在 G 上一定可以取到最小值 m , 与最大值 M , 且最小值 m 与最大值 M 的点一定包含在区域内部驻点, 内部偏导数不存在点, 边界上最大值与最小值之中.

计算出所有这怀疑点的函数值, 最大者为区域上的最大值, 最小者为区域上的最小值

求 $u=f(x,y,z)$ 在满足条件 $\varphi(x,y,z)=0$ 下的极值, 称为条件极值. 设 $P_0(x_0,y_0)$ 处取到极值

$$\varphi(x,y,z)=0 \text{ 确定 } z=z(x,y) \quad \frac{\partial z}{\partial x} = \frac{-\varphi'_x}{\varphi'_z}$$

$$\frac{\partial z}{\partial y} = -\frac{\varphi'_y}{\varphi'_z} \quad (\varphi'_z \neq 0) \quad \frac{\partial u}{\partial x} = f'_x + f'_z \cdot \frac{\partial z}{\partial x}$$

$$\frac{\partial u}{\partial x} = f'_x + f'_z \cdot \frac{\partial z}{\partial x} = f'_x - f'_z \cdot \frac{\varphi'_x}{\varphi'_z} = 0$$

$$\frac{\partial u}{\partial y} = f'_y + f'_z \cdot \frac{\partial z}{\partial y} = f'_y - f'_z \cdot \frac{\varphi'_y}{\varphi'_z} = 0$$

$$\frac{f'_x}{\varphi'_z} = \frac{f'_z}{\varphi'_z} \quad \frac{f'_y}{\varphi'_z} = \frac{f'_z}{\varphi'_z} \quad \therefore \frac{f'_x}{\varphi'_x} = \frac{f'_y}{\varphi'_y} = \frac{f'_z}{\varphi'_z} = -\lambda$$

称 λ 为参数, 则 $P_0(x,y)$ 满足:

$$\begin{cases} f'_x + \lambda \varphi'_x = 0 \\ f'_y + \lambda \varphi'_y = 0 \\ f'_z + \lambda \varphi'_z = 0 \\ \varphi(x,y,z) = 0 \end{cases} \quad \begin{aligned} &\text{令 } L(x,y,z,\lambda) = f(x,y,z) + \lambda \varphi(x,y,z) \\ &\text{是 } L(x,y,z,\lambda) \text{ 的驻点} \end{aligned}$$

总结: 1. $L(x,y,z,\lambda) = f(x,y,z) + \lambda \varphi(x,y,z)$

$$2. \text{ 求 } \begin{cases} L'_x = f'_x + \lambda \varphi'_x = 0 \\ L'_y = f'_y + \lambda \varphi'_y = 0 \\ L'_z = f'_z + \lambda \varphi'_z = 0 \\ L'_\lambda = \varphi(x,y,z) = 0 \end{cases} \quad \text{求出 } (x_0, y_0, z_0)$$

3. 计算出函数值.

这个方法, 我们称为拉格朗日乘数法

例: 求 $u = \sqrt{x^2+y^2+z^2}$ 在满足 $\varphi(x,y,z)=0$ 条件的最大与最小值

$$\text{解: } L(x,y,z,\lambda) = \sqrt{x^2+y^2+z^2} + \lambda \varphi(x,y,z)$$

$$L'_x = \frac{x}{\sqrt{x^2+y^2+z^2}} + \lambda \varphi'_x = 0 = L'_y = L'_z = L'_\lambda = 0$$

$$\text{解法: 令 } L(x,y,z,\lambda) = x^2+y^2+z^2 + \lambda \varphi(x,y,z)$$

$$L'_x = 2x + \lambda \varphi'_x = 0 \quad L'_y = L'_z = \varphi(x,y,z) = 0$$

注: $\sqrt{x^2+y^2+z^2}$ 与 $x^2+y^2+z^2$ 在同-处取最大(小)

例: 求 $u=f(x,y,z)$ 在满足 $\varphi(x,y,z)=0$.

$\varphi(x,y,z)=0$ 条件下的极值.

$$\text{解: 令 } L(x,y,z,\lambda,\mu) = f(x,y,z) + \lambda \varphi(x,y,z) + \mu \psi(x,y,z)$$

$$\begin{cases} L'_x = f'_x + \lambda \varphi'_x + \mu \psi'_x = 0 \\ L'_y = f'_y + \lambda \varphi'_y + \mu \psi'_y = 0 \\ L'_z = f'_z + \lambda \varphi'_z + \mu \psi'_z = 0 \\ L'_\lambda = \varphi(x,y,z) = 0 \\ L'_\mu = \psi(x,y,z) = 0 \end{cases}$$

例: 求 $z = x^2+y^2+2x+y+1$ 在 $x^2+y^2 \leq 1$ 上的最大值与最小值.

解: 由 $x^2+y^2 \leq 1$ 是有界闭区域, 且该函数在该有界闭区域上连续, 因此一定存在最大值 M 与最小值 m .

$$1) \text{ 求 } \begin{cases} \frac{\partial z}{\partial x} = 2x+2 = 0 \\ \frac{\partial z}{\partial y} = 2y+1 = 0 \end{cases} \quad \begin{cases} x = -1 \\ y = -\frac{1}{2} \end{cases}$$

由于不在内部, 故舍去.





(2) 内部无偏导数不存在点

(3) 求 $z = x^2 + y^2 + 2x + y + 1$ 在满足 $x^2 + y^2 - 1 = 0$ 的条件极值.

$$z = 2x + y + 2$$

$$\text{令 } L(x, y, \lambda) = 2x + y + 2 + \lambda(x^2 + y^2 - 1)$$

$$\begin{cases} L'_x = 2 + 2\lambda x = 0 & \lambda = -\frac{1}{x} \Rightarrow x = 2\lambda \\ L'_y = 1 + 2\lambda y = 0 & \lambda = -\frac{1}{2y} \\ L'_\lambda = x^2 + y^2 - 1 = 0 \end{cases}$$

$$\text{解得 } \begin{cases} x = \frac{2}{\sqrt{5}} \\ y = \frac{1}{\sqrt{5}} \end{cases} \quad \begin{cases} x = -\frac{2}{\sqrt{5}} \\ y = -\frac{1}{\sqrt{5}} \end{cases}$$

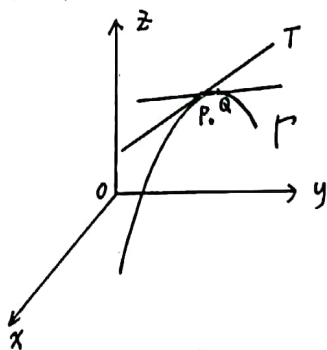
$$z\left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right) = 2 - \sqrt{5}$$

$$z\left(-\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right) = 2 + \sqrt{5}$$

$$\therefore m = 2 - \sqrt{5} \quad M = 2 + \sqrt{5}$$

偏导数在几何上的应用.

1. 求空间曲线上一点的切线与法平面



设曲线 Γ $\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$ 参数 $t = t_0$ 对应曲线上点 $P_0(x_0, y_0, z_0)$

处切线 P_0T 的方程.

$$x_0 = x(t_0) \quad y_0 = y(t_0) \quad z_0 = z(t_0)$$

$$\text{设 } Q(x(t_0 + \Delta t), y(t_0 + \Delta t), z(t_0 + \Delta t))$$

$$Q \rightarrow P_0 \Leftrightarrow \Delta t \rightarrow 0. \quad (x(t), y(t), z(t) \text{ 在 } t_0 \text{ 处连续})$$

$$\text{割线 } P_0Q \text{ 的方向向量 } \vec{P_0Q} = \{\Delta x, \Delta y, \Delta z\}$$

$$\text{有 } \left\{ \frac{\Delta x}{\Delta t}, \frac{\Delta y}{\Delta t}, \frac{\Delta z}{\Delta t} \right\} \text{ 是 } P_0Q \text{ 的方向向量.}$$

$$\left(\lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{x(t_0 + \Delta t) - x(t_0)}{\Delta t} = x'(t_0) \right)$$

要求 $x'(t_0), y'(t_0), z'(t_0)$ 均存在, 且不同时为 0. 令 $\Delta t \rightarrow 0$, P_0Q 方向向量的极限

$\{x'(t_0), y'(t_0), z'(t_0)\}$ 是切线 P_0T 的方向向量

且切线 P_0T 方程为:

$$\frac{x - x(t_0)}{x'(t_0)} = \frac{y - y(t_0)}{y'(t_0)} = \frac{z - z(t_0)}{z'(t_0)}$$

$$\text{即 } \frac{x - x_0}{x'(t_0)} = \frac{y - y_0}{y'(t_0)} = \frac{z - z_0}{z'(t_0)}$$

在 P_0 点的法平面为:

$$x'(t_0)(x - x_0) + y'(t_0)(y - y_0) + z'(t_0)(z - z_0) = 0$$

$\pm \{x'(t_0), y'(t_0), z'(t_0)\}$ 称为切线的方向向量.

例: 求螺旋线 $x = a \cos t, y = a \sin t, z = vt$

(a, v 均不为 0, 为常数) 求 $t = \frac{\pi}{3}$ 时曲线上点的切线方程与法平面方程

解: 当 $t = \frac{\pi}{3}$ 时, 对应的点 $x_0 = a \cos \frac{\pi}{3} = \frac{1}{2}a$

$$y_0 = a \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}a \quad z_0 = \frac{\pi}{3}v$$

$$x'(t) = -a \sin t \quad y'(t) = a \cos t \quad z'(t) = v$$

$$\text{切线的方向向量 } \vec{v} = \left\{ -\frac{\sqrt{3}}{2}a, \frac{1}{2}a, v \right\}$$

$\therefore$ 在 $t = \frac{\pi}{3}$ 对应曲线上点 P_0 处的切线方程

$$\text{为: } \frac{x - \frac{1}{2}a}{-\frac{\sqrt{3}}{2}a} = \frac{y - \frac{\sqrt{3}}{2}a}{\frac{1}{2}a} = \frac{z - \frac{\pi}{3}v}{v}$$

法平面方程:

$$-\frac{\sqrt{3}}{2}a(x - \frac{1}{2}a) + \frac{1}{2}a(y - \frac{\sqrt{3}}{2}a) + v(z - \frac{\pi}{3}v) = 0$$

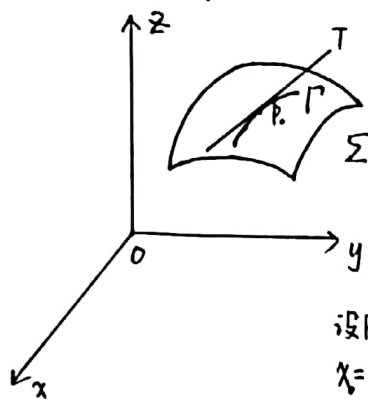




曲面上一点的切平面与法线方程

设 Σ 为空间一块曲面 $P_0(x_0, y_0, z_0) \in \Sigma$

Σ 的方程为 $F(x, y, z) = 0$



$$F(x_0, y_0, z_0) = 0$$

$$\forall \Gamma \subset \Sigma, P_0 \in \Gamma$$

$$\therefore \text{设 } \Gamma: \begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$$

设 P_0 对应的参数为 $t = t_0$

$$x_0 = x(t_0), y_0 = y(t_0), z_0 = z(t_0)$$

设 $x'(t_0), y'(t_0), z'(t_0)$ 都存在且不同时为 0

在 P_0 处曲线 Γ 的切线的方向向量 $\vec{v} = \{x'(t_0), y'(t_0), z'(t_0)\}$

由 $\Gamma \subset \Sigma$ 得到 $F(x(t), y(t), z(t)) \equiv 0, t \in D$

(要求 $F(x, y, z)$ 在 $U(P_0)$ 内具有连续的一阶偏导数且

$x'(t), y'(t), z'(t)$ 在 t_0 的邻域内存在)

(1) 两边对 t 求导:

$$F'_x(x(t), y(t), z(t)) \cdot x'(t) + F'_y(x(t), y(t), z(t)) \cdot y'(t) + F'_z(x(t), y(t), z(t)) \cdot z'(t) = 0$$

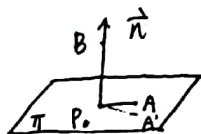
为了得到 (2) 式, 只须 F'_x, F'_y, F'_z 在 P_0 处具有连续的一阶偏导数⁽²⁾

且不同时为 0, $x'(t_0), y'(t_0), z'(t_0)$ 存在, 不同时为 0, 有

把 (1) 中的 t 有 $F'_x(P_0)x'(t_0) + F'_y(P_0)y'(t_0) + F'_z(P_0)z'(t_0) = 0$ 即

$$\{x'(t_0), y'(t_0), z'(t_0)\} \perp \{F'_x(P_0), F'_y(P_0), F'_z(P_0)\}$$

$\vec{n} \perp$ 平面 $\pi, P_0 \in \pi$



若 $P_0A \perp \vec{n}$, 则 $P_0A \subset \pi$

证: 平面 BP_0A 与 π 的交线设为 P_0A'

$\angle BPA' = \frac{\pi}{2} \because A$ 与 A' 共面且

$P_0A \perp P_0B, P_0A' \perp P_0B \therefore A$ 与 A' 重合

$\therefore P_0A \subset \pi$

过 P_0 点, 曲线的切线都在 π 经过 P_0 , 法向量为 $\{F'_x(P_0), F'_y(P_0), F'_z(P_0)\}$

的平面上, 该平面称为在曲面 Σ 上点 P_0 处的切平面.

切平面的方程为:

$$F'_x(P_0)(x-x_0) + F'_y(P_0)(y-y_0) + F'_z(P_0)(z-z_0) = 0$$

$\pm \{F'_x(P_0), F'_y(P_0), F'_z(P_0)\}$ 称为曲面在 P_0 处的法向量

法线方程为:

$$\frac{x-x_0}{F'_x(P_0)} = \frac{y-y_0}{F'_y(P_0)} = \frac{z-z_0}{F'_z(P_0)}$$

若曲面 Σ 方程为 $z = f(x, y)$, 求曲面上点

$P_0(x_0, y_0, z_0)$ ($z_0 = f(x_0, y_0)$) 处的切平面与法线方程.

解: $F(x, y, z) = f(x, y) - z = 0$

$$F'_x = f'_x(x, y) \quad \text{外函数对中间变量 } x \text{ 偏导}$$

$$F'_y = f'_y(x, y)$$

$$F'_z = -1$$

只需 $z = f(x, y)$ 在 (x_0, y_0) 处具有连续的一阶偏导数, 此时法向量

$$\vec{n} = \{f'_x(x_0, y_0), f'_y(x_0, y_0), -1\}$$

从而, 在曲面 Σ 上 $P_0(x_0, y_0, z_0)$ 处

切平面方程为

$$f'_x(x_0, y_0)(x-x_0) + f'_y(x_0, y_0)(y-y_0) - (z-z_0) = 0$$

法线方程为

$$\frac{x-x_0}{f'_x(x_0, y_0)} = \frac{y-y_0}{f'_y(x_0, y_0)} = \frac{z-z_0}{-1}$$





设曲线 $\Gamma \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$

求曲线上一点 $P_0(x_0, y_0, z_0)$ 处的
切线与法平面方程

切线方程为 $\begin{cases} \text{曲面 } F \text{ 在 } P_0 \text{ 点的切平面方程} \\ \text{曲面 } G \text{ 在 } P_0 \text{ 点的切平面方程} \end{cases}$

法平面方程为

切线的方向向量 $\vec{v} = \{F'_x(P_0), F'_y(P_0), F'_z(P_0)\} \times \{G'_x(P_0), G'_y(P_0), G'_z(P_0)\}$

法平面经过 P_0 , 法向量为 $\vec{v}$, 用点法式求出法平面方程.

以 $z = f(x, y)$ 在 $P_0(x_0, y_0)$ 为例.

$z = f(x, y)$ 在 $P_0(x_0, y_0)$ 处沿任何方向的方向导数存在.

(方向导数存在的充分条件)

(可微的必要条件)

(可微的
充分条件)

$f'_x(x, y)$
 $f'_y(x, y)$
在 $P_0(x_0, y_0)$ 处连续

可微 $\iff f'_x(x_0, y_0)$
 $f'_y(x_0, y_0)$ 存在.
连续

